

Fig. 3: Actual-size PCB layout of adjustable power supply

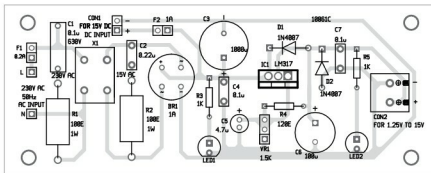


Fig. 4: Components layout for the power supply PCB

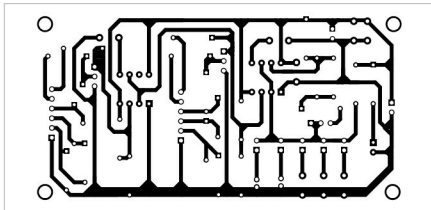


Fig. 5: Actual-size PCB layout of the signal generator

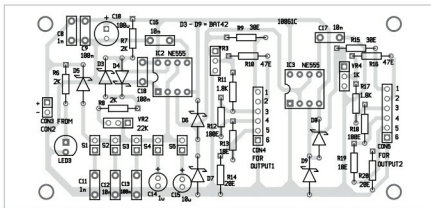


Fig. 6: Components layout for the signal generator PCB

PARTS LIST

Semiconductors:

- IC1 - LM317 adjustable voltage regulator
- IC2, IC3 - NE555 timer
- D1, D2 - 1N4007 rectifier diode
- D3-D9 - BAT42 Schottky diode
- LED1-LED3 - 5mm LED
- BR1 - 1A bridge rectifier

Resistors (all 1/4-watt, 5% carbon, unless stated otherwise):

- R1, R2 - 100-ohm, 1-watt
- R3, R5 - 1-kilo-ohm
- R4 - 120-ohm
- R6-R8 - 2-kilo-ohm
- R9, R15 - 30-ohm
- R10, R16 - 47-ohm
- R11, R17 - 1.8-kilo-ohm
- R12, R18 - 180-ohm
- R13, R19 - 18-ohm
- R14, R20 - 20-ohm
- VR1 - 1.5-kilo-ohm potmeter
- VR2 - 22-kilo-ohm potmeter
- VR3, VR4 - 1-kilo-ohm potmeter

Capacitors:

- C1 - 0.1μF, 630V polyester
- C2 - 0.22μF ceramic disk
- C3 - 1000μF, 35V electrolytic
- C4 - 0.1μF ceramic disk
- C5 - 4.7μF, 35V electrolytic
- C6, C10 - 100μF, 35V electrolytic
- C7 - 0.1μF ceramic disk
- C8, C11 - 1nF ceramic disk
- C9, C13, C18 - 100nF ceramic disk
- C12, C16, C17 - 10nF ceramic disk
- C14 - 1μF, 35V electrolytic
- C15 - 10μF, 35V electrolytic

Miscellaneous:

- CON1, CON3 - 2-pin connector
- CON2 - 2-pin terminal connector
- CON4, CON5 - 6-pin connector
- S1-S5 - On/off switch
- F1 - 0.2A fuse with holder
- F2 - 1A fuse with holder
- X1 - 230V AC primary to 15V, 500mA secondary transformer
- 15V DC regulated supply or 230V AC mains
- Heat sink for IC1

An actual-size PCB layout of the signal generator circuit (Fig. 2) is shown in Fig. 5 and its components layout in Fig. 6. After assembling the circuit on the PCB, connect variable power supply using a two-wire cable from CON2 to CON3. Connect LED3, switches S1 through S5 and potmeters VR2 through VR4 on the front panel for power-status indication, frequency selection and signal amplitude control, respectively.

Note. For testing, you can also use 6V, 9V or 12V DC supply at CON3. **EFY**

Petre Tzvetz Petrov was a researcher and assistant professor in Technical University of Sofia, Bulgaria and expert lecturer in OFPPT(Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria

